Liquid Staking: When Does It Help?

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Abstract

Liquid staking, which allows to redeploy capital from a blockchain's staking pool into decentralized finance (DeFi), is a recent instance of a financial innovation promoted for its "capital efficiency". But is it always desirable? We build a tractable general equilibrium setup to answer this question, where for concreteness the DeFi outlet is a lending pool. In the benchmark case without friction, liquid staking does improve upon traditional staking. But we prove that liquid staking can backfire when a friction is at play. (i) When investors are large, liquid staking exacerbates concentration issues, so that replacing traditional by liquid staking may be inefficient. (ii) When the security of the blockchain is endogenous, liquid staking can worsen security.

Keywords: blockchain, decentralized finance, Proof-of-Stake, liquid staking, wrapped tokens, welfare, large agents, market concentration, security.
JEL Classification: D47, G29, O3.

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1 Introduction

"In today's capital markets, securities seldom lie idle."

Bank for International Settlements (1999), report on securities lending transactions.

The notion of capital efficiency regularly motivates financial innovation. A case in point is the keen interest met by mechanisms such as *liquid staking* and *restaking* in decentralized finance (DeFi). In a Proof-of-Stake (PoS) blockchain such as Ethereum, liquid staking consists in entrusting native tokens to validators instead of locking them directly in the staking pool; validators then assume the responsibility of participating in the PoS consensus mechanism. Each delegated token is represented by a 'wrapped' token which functions as a certificate of deposit and, crucially, becomes an asset in its own right. In particular, it can be redeployed onto DeFi platforms. That is, capital which was previously idle in the staking pool is assigned a new purpose.

Figure 1: Liquid staking on Ethereum.
Source: DefiLlama.

The positives of liquid staking have been abundantly highlighted. A Kraken post emphasizes the usefulness of this mechanism to reduce illiquidity and "improve capital efficiency".[1] Going further, the Swiss bank Amina states that liquid staking is "solving the capital lock-up

[1] https://www.kraken.com/learn/what-is-liquid-staking, Dec. 6, 2024.

problem."[2]

In this paper, we aim at clarifying the "capital efficiency" intuition: what exactly makes liquid staking preferable to traditional staking (which we will also call 'direct staking'), and is it even the case that it is always preferable? We begin with a simple benchmark model which does deliver a "capital efficiency" result: we prove that, in the absence of frictions, liquid staking dominates traditional staking. Yet, this leaves open the questions of whether the capital efficiency logic would work the same way in the presence of realistic frictions and if the conclusion that liquid staking is desirable still holds in this context. One well-identified "friction" is that liquid staking may mechanically entail additional risks, as it involves delegation and hence potential losses.[3] Obviously, our point in the paper is not to verify that liquid staking is undesirable when these costs are large—which is mechanical. The question we answer is radically different: starting with a configuration where liquid staking is better, can the introduction of a realistic friction *at play regardless of the chosen staking type* render traditional staking in fact better? We answer these questions by successively studying the impact of the two following frictions: the existence of large investors (Friction 1), and the endogeneity of blockchain security (Friction 2).

In traditional finance, the importance of large agents is well-recognized, as evidenced by a surge in academic attention over the recent years (see, e.g. the survey by Rostek and Yoon (2025)). Market concentration also matters in decentralized finance. For instance, 95% of ETHs are held on 0.3% of the wallets, with the ETH distribution resembling wealth distribution in traditional economies (Celig, Ockenga, and Schoder (2025)); and the existence of large agents with the ability to impact crypto prices (the so-called 'whales') is established (see e.g. Chernoff and Jagtiani (2024)). Further, a Global Financial Stability Report from the IMF finds that "DeFi seems to be largely used by a small number of institutional entities" (International Monetary Fund (2022)). This motivates the study of Friction 1.

[2] https://aminagroup.com/research/the-current-state-of-staking-institutional-adoption-at-scale

[3] Due to validator carelessness or downright misbehaviour (see, e.g. Lido (2020), Tzinas and Zindros (2023)). We capture this with a "delegation costs" parameter.

Studying Friction 2 is equally natural: if blockchain security was automatically guaranteed, there would be no staking in the first place. It is thus important to take into account the fact that only a staking of large enough value makes the blockchain resilient to attacks.

Building on Carré and Gabriel (2022), we construct a general equilibrium framework to compare the efficiency of liquid versus direct staking in the various environments described above. Overlapping generations of *investors* and *DeFi users* interact. Investors decide whether to use part of their real good endowment to purchase tokens—the blockchain's native currency. These tokens can generate return in two ways: by being staked, or by being deployed in DeFi. For concreteness, we consider that the DeFi outlet is a lending pool. When staking is liquid, investing one token exposes to both sources of return, as the wrapped asset obtained after delegation is then redeployed in DeFi. On the demand side, DeFi users want to borrow to finance consumption. In all the considered environments, a welfare metrics, taking into account the utilities of the various agents, appears naturally. We can thus compare the efficiency of both staking types. The benchmark model features no other friction than the delegation costs associated with liquid staking. The model for Friction 1 (large agents) considers a finite numbers of investors, instead of a mass 1 continuum. The model for Friction 2 (endogenous security) takes into account the link between the market value of the staking pool and the resilience of blockchain to attacks. This resilience becomes an equilibrium object that impacts welfare. Indeed, following an attack, DeFi users are barred from borrowing and have minimal utility.

Our main results are the following.

First, we formalize the "capital efficiency" intuition. In the benchmark model we prove that, as long as the liquid staking delegation costs are not too large, liquid staking dominates direct staking in terms of efficiency. Specifically, we show that liquid staking is weakly better when the borrowing cost on the DeFi outlet is fixed, and strictly better when the DeFi outlet design is adjusted optimally. In the model, the exact channel through which capital efficiency is at play is that under liquid staking, investors are ready to accept a lower return from the

DeFi outlet, because they can compound it with the staking return. This lowers DeFi users' borrowing costs and improves welfare. As expected, liquid staking is desirable because it allows to compound investments. But while this logic works well in a frictionless environment, our next results evidence that, when a friction prevails, liquid staking can backfire in that this very ability to compound may generate worse outcomes than under direct staking.

Second, we study the case where investors are large in that their aggregate initial endowment is split into a finite number of players n, instead of a continuum. We find that there are configurations where liquid staking was desirable in the perfect competition case, but gets dominated by direct staking as agents are large. Specifically, we prove that there exists a range of intermediate delegation costs such that the preferred staking type is direct for small n and liquid for large n. The key driver of this result is that liquid staking prompts a larger increase of the investors' required DeFi rate of return, compared to direct staking. This harms DeFi users. Concentration increases the required rate of return on blockchain-DeFi as an extra unit invested earns this return minus a price impact term, which needs to amount to at least the outside option rate. Stronger concentration implies a larger price impact term and thus increases the equilibrium blockchain-DeFi returns. But our main insight is that this increase is *larger under liquid staking*, for two reasons. Under direct staking, the "markets" for staking and lending are separate, and the extra return on staking commanded by concentration comes from a downward adjustment of the staking level. This adjustment is impossible under liquid staking—as the entire token supply is staked and redeployed—and investors get their desired level of compounded return by requiring a larger DeFi rate. Further, we compare our model outcome to the cases where there is concentration in one market only (staking or lending), and establish that concentration has a linear impact on borrowing rates when staking is direct, but a superlinear impact when staking is liquid. That is, under direct staking the adjustments to the required DeFi return add up across markets where there is concentration, while under liquid staking they more than add up. Taken together, these results indicate that the "virtues of compounding" associated with liquid staking come with an undesirable byproduct, namely that compounding investments "imports" price impact from other markets and makes price impacts also compound.

Third, we consider another friction, namely that the resilience of the blockchain to attacks is an increasing function of the value of its staking pool—rather than assuming that the blockchain necessarily remains forever in operation. We prove that, even absent any liquid staking delegation cost, when the borrowing rate on the DeFi outlet is fixed, liquid staking can decrease welfare. Here, compounding backfires because it increases the return that can be expected from blockchain-DeFi and thus makes some degree of blockchain attack risk compatible with equilibrium. Intuitively, if investors require a 3% expected return to invest in the token, but observe a 10% announced return in blockchain-DeFi, they may think that the extra 7% are a compensation for attack risk. This opens the possibility of a self-fulfilling risky equilibrium. However, this undesirable effect can be eliminated by an adequate adjustment of the DeFi rate : we prove that, when DeFi rates are adjusted optimally and as long as delegation costs are not too large, liquid staking is more efficient than direct staking.

Related works. Blockchain economics started with the analysis of consensus layers. A non-exhaustive list includes Biais, Bisière, Bouvard, and Casamatta (2019), Prat and Walter (2021), Huberman, Leshno, and Moallemi (2021), Pagnotta (2022) for Proof-of-Work; and John, Rivera, and Saleh (2020, 2022), Saleh (2021), Rosu and Saleh (2021), Cong, He, and Tang (2025)[4] for Proof-of-Stake.

The appearance of DeFi platforms built upon the consensus layer then fostered the development of an additional branch of the literature: the behaviour and strategies of DeFi actors are studied e.g. in Lehar and Parlour (2022), Park and Stinner (2023), Heimbach, Schertenleib, and Wattenhofer (2023), Barbon, Barthélemy, and Nguyen (2023), Heimbach and Huang (2023), and Cornelli, Gambacorta, Garratt, and Reghezza (2024).

How to efficiently design DeFi lending platforms, and notably their interest rate mechanism (IRM) is studied in Carré and Gabriel (2022), Rivera, Saleh, and Vandeweyer (2023),

[4]While Cong, He, and Tang, focus on traditional staking, they mention liquid staking as an important research avenue, that may be considered in an extension of their model.

Cohen, Sánchez-Betancourt, and Szpruch (2023) and Bertucci, Bertucci, Gontier-Delauney, Guéant, and Lesbre (2024). Chiu, Ozdenoren, Yuan, and Zhang (2024) identify a price-liquidity feedback mechanism in DeFi lending, in a model that focuses on collateral quality rather than on IRMs.

The empirics of liquid staking markets are studied in several articles. Scharnowski and Jahanshahloo (2024) provide an econometric analysis of the drivers of liquid staking token prices. Gogol, Kraner, Schlosser, Yan, Tessone, and Stiller (2024), Gogol, Velner, Kraner, and Tessone (2024) and Grandjean, Heimbach, and Wattenhofer (2023) provide a descriptive and operational view of the liquid staking market. While we are interested in the deployment of liquidity on DeFi lending protocols—and thus on the design of wrapped tokens lending pools— Gogol, Fritsch, Schlosser, Messias, Kraner, and Tessone (2024) consider an important alternative: liquid staking for deployment on automated market makers. In operational research-oriented approaches, Xiong, Wang, and Wang (2024) and Xiong, Wang, Chen, Knottenbelt, and Huth (2024) explore the profitability and risks associated with various strategies in the liquid staking market. Jermann (2024) performs a welfare analysis of an economy that uses a Proof-of-Stake (crypto)currency, using a DSGE model. He takes liquid staking into account in his data and estimation, but does not examine its contrast with direct staking. Drissi, Feinstein, and Williams (2026) also provide a macro-finance framework to analyze optimal token inflation and evidence that this policy instrument may be lost when liquid staking is introduced.

2 The Model

2.1 Model Description

The model builds on Carré and Gabriel (2022). We consider a Proof-of-Stake blockchain upon which one DeFi lending pool operates: (i) under liquid staking (LS), all tokens are staked and the resulting wrapped token is redeployed in the lending pool, (ii) under direct staking (DS), tokens are either staked or lent.

This section describes the LS model without friction. The end of this section 2.1, explains the necessary adjustments for the case of DS. Sections 4 and 5 present the adjustments necessary to analyze the two frictions considered in the paper, and the corresponding results.

Time is discrete, $t = 0, 1, 2 \ldots$. Each period t is divided into two subperiods, denoted $\underline{t}$ and $\bar{t}$. There is a real consumption good and a blockchain-operated currency (native token). A *wrapped token* is issued whenever a token is delegated for staking: it functions as a certificate of deposit representing the delegated native token and the cumulative Proof-of-Stake rewards that it generated, net of potential costs that we introduce later.

The native token is in supply Ω_t and has (endogenous) price P_t, in real good units, in a centralized market (denoted CM) which opens at the beginning of each period $\underline{t}$.

The only way to transfer the consumption good across periods is to invest it at time $\underline{t}$ into a risk-free technology (independent of the blockchain environment), which has gross return $r > 1$ between $\underline{t}$ and $\underline{t+1}$. If not, the good perishes at the end of period t.

There are three types of agents: *investors*, DeFi *users* and *sellers*.[5] There is a mass 1 continuum of each kind. (Section 4 analyzes the case where there is a finite number n of investors.) All agents live over three subperiods: generation t is in existence at $\underline{t}$, $\bar{t}$ and $\underline{t+1}$. No discount rate is applied to periods $\bar{t}$ and $\underline{t+1}$.

Investors are born with a large endowment in the real good and are risk-neutral, $U_{\mathrm{I}} = c_{\underline{t}} + c_{\bar{t}} + c_{\underline{t+1}}$. Given that $r > 1$, investors simply maximize $\mathbb{E}_{\underline{t}}[c_{\underline{t+1}}]$. They do so by choosing what share of their endowment to invest into the blockchain environment, i.e. how many tokens to purchase, liquid stake and redeploy onto the DeFi lending pool.[6] The return on investing on the blockchain comes from the following elements.

[5] Omitting the validators, which are passive in the model.

[6] The remainder of their endowment being invested in the risk-free technology. Conditional on investing into the blockchain, it is straightforward to see that liquid staking and redeploying is indeed optimal.

First, liquid staking alone implies a gross yield of

$$\gamma_t \equiv (1-\alpha)\left(1+\frac{R_t}{S_t^{liq}}\right). \tag{1}$$

In this expression, $\alpha \in [0;1)$ represents the fraction of funds absorbed by delegation costs, R_t the Proof-of-Stake reward, and S_t^{liq} the size of the staking pool. The investor receives the standard staking return of $1+\frac{R_t}{S_t^{liq}}$, net of delegation costs. We let

$$\rho = \frac{\Omega_t + R_t}{\Omega_t} \tag{2}$$

denote the inflation rate, which we assume to be constant.

Second, redeployment of the wrapped token on the lending pool guarantees a return of $1+u_t y$, where $u_t = \frac{D_t}{S_t^{liq}}$ is the utilization rate, D_t is the amount borrowed by DeFi users, and the S_t^{liq} redeployed wrapped tokens are the pool's supply. y is the borrowing rate on the DeFi pool, and will be endogenized in our analysis.[7]

Since not all tokens are lent, the net return for lending is $u_t y$.

Finally, the nominal returns above are to be discounted by the inflation rate ρ. Hence, the gross return on the blockchain investment is

$$\gamma_t(1+u_t y)\rho^{-1}. \tag{3}$$

DeFi users are only able to produce the consumption good at $\underline{t+1}$, with a production technology which outputs one unit of good for each unit of effort, but desire to consume at time t: their utility is $V = v(Q_t) - e_{\underline{t+1}}$, where v is assumed to be CRRA with parameter denoted $\frac{1}{k}$. Q_t is their time-t consumption, derived from their real good purchases, which they finance via DeFi borrowing. $e_{\underline{t+1}}$ is the production effort they make to be able to repay

[7]That is, we consider constant interest rate mechanisms (IRMs). This is without loss of generality since, as shown in Carré and Gabriel (2022) such IRMs are sufficient to implement efficient outcomes when parameters are known, which is the case here. How parameter uncertainty impacts IRM design has already been studied in the literature and is not the focus of the current paper.

their debt.[8] Users owe $D_t(1+y)$ units of the wrapped token, which represent $\gamma_t D_t(1+y)$ native tokens, and thus their production effort is

$$e_{\underline{t+1}} = \gamma_t P_{t+1} D_t (1+y). \tag{4}$$

Sellers are born with no endowment, can produce the consumption good at $\bar{t}$ (with the same technology as users), and value consumption at $\underline{t+1}$. Their utility is $U_{\mathrm{S}} = -e_{\bar{t}} + c_{\underline{t+1}}$. They have no bargaining power, i.e. the payment they demand from DeFi users for their time-t production Q_t allows them to consume $c_{\underline{t+1}}$ such that they break even, $U_{\mathrm{S}} = 0$. One wrapped token obtained at $\bar{t}$ being worth $\gamma_t P_{t+1}$ units of good in the next CM, we have

$$Q_t = e_{\bar{t}} = \gamma_t P_{t+1} D_t. \tag{5}$$

Having linked the quantity Q_t consumed by the DeFi users with the quantity D_t they borrowed, we can write the DeFi users maximisation program as:

$$\max_{D_t} v(\gamma_t P_{t+1} D_t) - \gamma_t P_{t+1} D_t (1+y). \tag{6}$$

In the frictionless model presented here, the price of the native token next period, P_{t+1}, is known since there is no risk and because we will focus on stationary equilibria. In Section 5, the blockchain can be attacked with a probability that depends on the market value of its staking pool, one of the consequences being that P_{t+1} becomes stochastic.

DS model. When staking is direct, there are two differences. First, investors have an additional maximization step, as, once entered into the blockchain, they must allocate their tokens between staking and lending. The former has a net return $\frac{R_t}{S_t}$, where S_t denotes the

[8]By selling the good they produced in the $t+1$-CM to obtain the amount of tokens they owe. See Carré and Gabriel (2022) for a way to model the collateral of DeFi users, making them able to borrow at t and finding it optimal to not part ways with their collateral.

aggregate quantity of staking; the latter has a net return $u_t y$ where $u_t = \frac{D_t}{L_t}$ and $L_t = \Omega_t - S_t$ is the aggregate quantity of lending. Second, since the native token does not bear the Proof-of-Stake interest, equation (5) becomes $e_{\bar{t}} = Q_t = P_{t+1} D_t$.

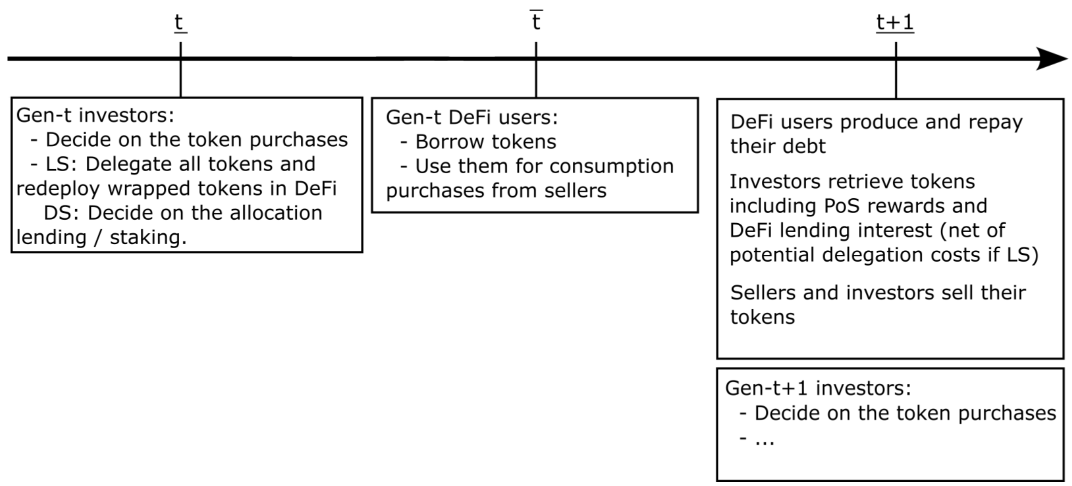

Figure 2: Timeline of the Benchmark Model

2.2 Equilibrium

As announced earlier, we focus on stationary equilibria where the total value of the native token supply is constant: $P_{t+1} = \rho^{-1} P_t$. First note that when investing into the blockchain is not sufficiently appealing, a trivial equilibrium where all funds are channeled into the risk-free project arises. (Would-be) DeFi users cannot borrow and their utility is minimal. We call this the *no-blockchain* equilibrium. Under liquid staking, when investors do enter, clearing of the CM implies that $S_t^{liq} = \Omega_t$ and $u_t = \frac{D_t}{\Omega_t}$. Thus, the *blockchain equilibrium* of our simple frictionless model can be reduced to the two equations:

$$r = (1 + u_t y)(1 - \alpha) \tag{L1}$$

$$D_t = \frac{1}{1-\alpha} \frac{(1+y)^{-k}}{P_t} \tag{L2}$$

(L1) expresses the indifference of the risk-neutral investors between entering and investing outside of the blockchain. It is obtained from (3), noting that $S_t^{liq} = \Omega_t$ and thus $\gamma_t \rho^{-1} = 1 - \alpha$. (L2) is the solution to the DeFi users' optimization problem (6).

We are effectively studying a static system. We can thus solve for time 0 and drop the dependency of all variables in the time index. Then, (P, D, u) is the time-0 equilibrium if and only if $(\rho^{-t} P, \rho^t D, u)$ is the time-t equilibrium.

Similarly, in the DS blockchain equilibrium clearing of the CM implies $\Omega_t = L_t + S_t$ and, with $u_t = \frac{D_t}{L_t}$, equilibrium is characterized by:

$$u_t y = \frac{R_t}{S_t} \tag{D3}$$

$$r = \frac{1 + u_t y}{\rho} \tag{D4}$$

$$D_t = \frac{\rho}{P_t} (1 + y)^{-k} . \tag{D5}$$

Rationed equilibria The systems of equations above characterize non-rationed equilibria, i.e. situations where the DeFi users' desired level of borrowing (given by the RHS of (L2) in the case of LS and the RHS of (D5) in the case of DS) is smaller or equal to the supply available on the lending pool. If not, DeFi users are rationed. Obviously, all our results and proofs account for the possibility that an equilibrium may be rationed or not. Non-rationed equilibria are the most important since, when rates are chosen optimally, rationed equilibria disappear.

2.3 Welfare

Without friction, both in LS and DS, the utilities of investors and sellers are constant; and in a blockchain equilibrium, DeFi users have utility

$$V = \frac{1}{k-1} (1 + y)^{1-k}, \tag{6}$$

in a no-blockchain equilibrium they have minimal utility. Thus welfare is minimal over no-blockchain equilibria and characterized by y over blockchain equilibria.[9]

3 Benchmark: "Capital Efficiency" of Liquid Staking

For LS, the utilization u in a blockchain equilibrium is directly obtained from (L1):

$$u = \frac{D}{\Omega} = \frac{1}{y}\left(\frac{r}{1-\alpha} - 1\right), \tag{7}$$

provided that the candidate value is in $[0;1]$. Otherwise the equilibrium is no-blockchain. This gives us the following necessary and sufficient condition for DeFi users to be able to borrow:

$$y \geq \underline{y}^{LS} \equiv \frac{r}{1-\alpha} - 1. \tag{8}$$

u and D are given by (7), and P then obtains immediately from (L2), completing the equilibrium description under LS.

For DS, similarly,

$$u = \frac{D}{L} = \frac{\rho r - 1}{y}, \tag{9}$$

so the blockchain equilibrium existence condition becomes

$$y \geq \underline{y}^{DS} \equiv \rho r - 1. \tag{10}$$

The equilibrium value of S is then given by (D3), which allows us to determine $D = uL = u(\Omega - S)$, pinning down P via (D5).

The equilibrium derivation above tells us that the DeFi rate y must be sufficiently ap-

[9]Under LS, there is an additional class of agents: validators. Since our main interpretation of the delegation cost is that validators can abscond with the native token and thus are malicious, it is natural to assign validators a zero weight in the welfare function. We also explored the case where the delegation cost represents a legitimate fee and validators' utility is included in welfare: this does not change our main results.

pealing for investors: if $y < \underline{y}^{LS}$ (resp. $y < \underline{y}^{DS}$) the equilibrium is no blockchain, i.e. investors channel their funds in their outside option. But it also helps finding out how the DeFi rate should optimally be adjusted depending on the staking regime. Indeed, since welfare decreases in y, $\underline{y}^{LS}$ and $\underline{y}^{DS}$ are the efficient DeFi rates under liquid and direct staking, respectively. Besides, from (7) and (9), at these efficient rates, the utilization is $u = 1$.

Having described the equilibrium under LS and DS, we can now state the main result of this section:

Proposition 1 ("capital efficiency")**.** *For sufficiently small delegation costs, i.e.* $\alpha < 1 - \frac{1}{\rho}$*:*
(i) At fixed y*, welfare under liquid staking is greater or equal to welfare under direct staking.*
(ii) Liquid staking strictly welfare-dominates direct staking when y *is adjusted optimally.*

(Proofs are relegated to Appendix A).

This result formalizes the "capital efficiency" view in the sense that, if there is no friction beyond the delegation cost associated with LS, then LS is more efficient than DS. Linearizing the upper bound $1 - \frac{1}{\rho}$ leads to the estimate that, for of a standard value of $\rho - 1$ around 3%, then LS is desirable as long as delegation costs are below 3%. The condition $\alpha < 1 - \frac{1}{\rho}$, or $\frac{1}{\rho} < 1 - \alpha$, is intuitive as it involves a comparison between measures of the cost of direct staking (ρ) and liquid staking (α). Consider indeed one unit of investment on the DeFi pool at some fixed rate. Under direct staking, the return of this investment needs to be corrected by inflation, implying a factor $\frac{1}{\rho}$. Under liquid staking, this factor does not exist, because the investment is simultaneously staked and hence, investors collect the staking reward which by construction exactly compensates for inflation; however, because of delegation costs, the return of the investment needs to be corrected by a factor $1 - \alpha$.

Figure 3 illustrates Proposition 1. In the left region, the equilibrium is no-blockchain both under LS and DS, and welfare is minimal. In the middle region, the equilibrium is blockchain under LS only, making LS strictly better than DS. In the right region, the equilibrium is blockchain both under LS and DS and since DeFi users face the same borrowing cost in both cases, LS and DS welfare coincide. Welfare at the optimally adjusted rate under LS is

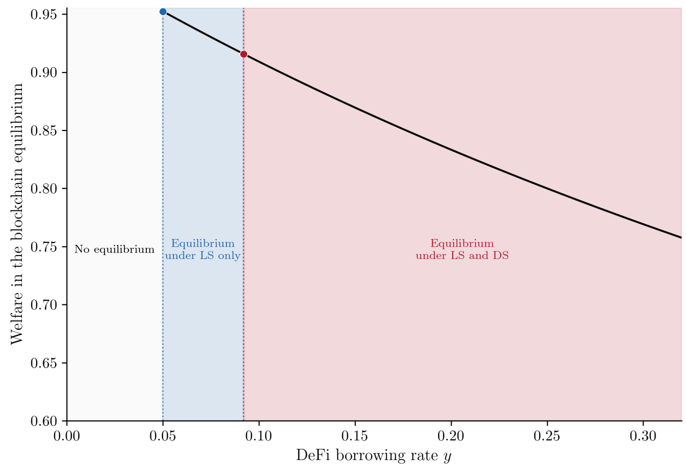

Figure 3: Welfare in the blockchain equilibrium under LS and DS
(Parameters used for the Figures are reported on the last page.)

visualized by the blue point at the left end of the black curve and welfare at the optimally adjusted rate under DS is visualized by the red point. As stated in the Proposition, the former is larger than the latter.

The Proposition and its graphical representation help us be more specific about how capital efficiency translates into larger welfare. At a fixed DeFi rate, LS may not lead to a strict improvement, but because LS allows investors to cumulate the returns from staking and lending (the capital is "efficient" in that it is used twice), it reduces the rate of return they require from the lending pool. This allows to offer better terms of borrowing to DeFi users and improves welfare.

The analysis so far assumes no friction (beyond the mechanical LS delegation costs). In the next two Sections, we show that when a friction prevails, LS may magnify it and end up being less efficient than DS.

4 Friction 1: Large Agents

We now depart from the benchmark model by assuming that the number n of investors is finite. We keep their total endowment e constant.[10]

[10] I.e. each investor starts with endowment e/n instead of $e\,\mathrm{d}i$.

The difference that it makes is that investors no longer take as given the amount of tokens that are liquid-staked. Instead, their own liquid-staking decision impacts the returns they obtain on the blockchain. Indeed, the net return on the staking activity is now $\frac{R}{s+S_{-}}$, where S_{-} is the quantity staked by others and s a given investor's staking; similarly, the net return on the lending activity is now $\frac{D}{s+S_{-}}y$. This leads to a new maximization program for investors, given and solved in Appendix A.3. The consequence is that only equation (L1) of the benchmark case is impacted, the system characterizing a (symmetric) equilibrium under liquid staking is now:

$$\frac{r}{1-\alpha} = \frac{1}{\rho}\left[(1+uy)\left[\frac{n-1}{n}\rho+\frac{1}{n}\right]\right]-\frac{1}{n}uy \tag{L11}$$

$$D = \frac{1}{1-\alpha}\frac{1}{P}(1+y)^{-k} \tag{L12}$$

Similarly, the equilibrium system under DS is:

$$r = \frac{1}{\rho}\left[1+\frac{(n-1)}{n}\frac{R}{S}\right] \tag{D13}$$

$$uy = \frac{R}{S} \tag{D14}$$

$$D = \frac{\rho}{P}(1+y)^{-k} \tag{D15}$$

$$\Omega = S+L \tag{D16}$$

Welfare is the sum of the utility of DeFi users and investors. When the equilibrium is no-blockchain, it is minimal.

We now characterize the equilibrium. Under LS, (L11) determines the value of u, which pins down $D = u\Omega$ and, ultimately, P, through (L12). With a similar solution in the DS case and denoting $c_n = \frac{n-1}{n} + \frac{1}{n\rho}$, we thus obtain:

Lemma 1. *(Equilibrium) Under liquid staking, the equilibrium is unique. It is blockchain* *for* $y \geq y_n^{LS} \equiv \frac{\frac{r}{1-\alpha}-c_n}{c_n-\frac{1}{n}}$ *and no-blockchain otherwise. When it is blockchain,*

$$u = \frac{\frac{r}{1-\alpha} - c_n}{y\left(c_n - \frac{1}{n}\right)} \tag{17}$$

$$P = \frac{1}{(1-\alpha)\Omega} \frac{\left(c_n - \frac{1}{n}\right)}{\left(\frac{r}{1-\alpha} - c_n\right)} y(1+y)^{-k}. \tag{18}$$

Under direct staking, the equilibrium is unique. It is blockchain for $y \geq y_n^{DS} \equiv \frac{n}{n-1}(r\rho - 1)$ *and no-blockchain otherwise. When it is blockchain,*

$$u = \frac{n}{n-1} \frac{r\rho - 1}{y} \tag{19}$$

$$P = \frac{\rho(n-1)y}{\Omega(n\rho(r-1) + \rho - 1)}(1+y)^{-k}. \tag{20}$$

Having determined the equilibrium both under LS and DS, we can determine explicitly the welfare of our economy and the efficient rates under both staking regimes.

Lemma 2. *(Welfare and optimal DeFi rates) With n investors, in equilibrium: (i) the welfare under liquid staking is*

$$W_n^{LS}(y) := re + \left[\left((1-r-\alpha)\frac{\left(c_n - \frac{1}{n}\right)}{(r - c_n(1-\alpha))} + 1 + \frac{1}{k-1}\right) y + \frac{1}{k-1}\right](1+y)^{-k}. \tag{21}$$

W_n^{LS} *decreases in y and thus under liquid staking the optimal DeFi rate is y_n^{LS}.*

(ii) the welfare under direct staking is

$$W_n^{DS}(y) := re + \left[\left(\frac{r\rho - 1}{n\rho(r-1) + \rho - 1} + \frac{1}{k-1}\right) y + \frac{1}{k-1}\right](1+y)^{-k}. \tag{22}$$

W_n^{DS} *decreases in y and thus under direct staking the optimal DeFi rate is y_n^{DS}.*

(When interested in comparing the two welfares, we will omit the common constant term re.) Note that, as in the benchmark case, from (17) and (19), at the efficient DeFi rates (respectively y_n^{LS} and y_n^{DS}), utilization is $u = 1$.

The efficient DeFi rates play an important role: they represent the rates of return (per unit effectively lent) required by the investors on the DeFi outlet to channel their funds into the blockchain. If these rates are large, borrowers' utility is harmed. Thus, one important step in understanding how the large agents friction impacts the relative efficiency of liquid vs direct staking—i.e. the quantity $W_n^{LS}(y_n^{LS}) - W_n^{DS}(y_n^{DS})$— is to characterize how it impacts y_n^{LS} and y_n^{DS}. To do so, rewrite the expression for y_n^{LS} found in Lemma 1 as:

$$y_n^{LS} = \frac{\overbrace{\frac{r}{1-\alpha} - 1}^{\text{frictionless } \underline{y}^{LS}} + \frac{\rho-1}{n\rho}}{1 - \frac{\rho-1}{n\rho} - \frac{1}{n}}. \tag{23}$$

The first insight of (23) is that concentration makes investors require larger DeFi rates. Indeed, y_n^{LS} decreases in n (and, as expected, it converges to the perfect competition benchmark $\underline{y}^{LS}$ as n becomes large). The fact that, in general, large agents make equilibrium returns larger is a simple consequence of the investors' first-order condition. Indeed they invest one additional unit up to the point where

$$\text{outside return} = \text{blockchain-DeFi return} - \text{impact on entire investment}. \tag{24}$$

For instance, in a DeFi pool with rate y and current utilization u, the profit of depositing one additional token can be decomposed as its own net return uy, minus the foregone return $|\Delta u|y$ on the entire investment, coming from the fact that the new deposit has induced a change in utilization $\Delta u < 0$. The same logic holds when considering the investment of one additional unit in the staking pool. Since the negative return impact is larger when agents are large, DeFi returns need to be larger in that case.

But (23) contains more information, as it also indicates the *contribution* of each "market"'s concentration to y_n^{LS}. Here we employ "market" loosely to refer to the two sources of investors' return on blockchain-DeFi: the staking pool and the lending pool. Since the n investors stake and redeploy, there is concentration in both markets. But we can switch off the impact of large agents in one market or the other by doing the following. To isolate the impact of concentration in the staking pool, we make a comparison with the model solution when the DeFi pool is cleared by a Walrasian auction (i.e. $u = 1$ and agents are price-takers in this pool). Let $y_{s,n}^{LS}$ be the equilibrium DeFi rate in this context. We have[11]

$$y_{s,n}^{LS} = \frac{\frac{r}{1-\alpha} - 1 + \frac{\rho-1}{n\rho}}{1 - \frac{\rho-1}{n\rho}}. \tag{25}$$

Hence, the correction $\Delta y_{s,n}^{LS} := y_{s,n}^{LS} - \underline{y}^{LS}$ to the frictionless rate $\underline{y}^{LS}$ brought by concentration in the staking market is exactly captured by the terms in blue in (23).

We can also isolate the impact of concentration in the lending pool by making a comparison with the model solution when the per-unit staking reward is fixed (instead of depending on the aggregate mass of staking). Let $y_{\ell,n}^{LS}$ be the optimal equilibrium DeFi rate in this context. We have

$$y_{\ell,n}^{LS} = \frac{\frac{r}{1-\alpha} - 1}{1 - \frac{1}{n}}. \tag{26}$$

Hence, the correction $\Delta y_{\ell,n}^{LS} := y_{\ell,n}^{LS} - \underline{y}^{LS}$ to the frictionless rate $\underline{y}^{LS}$ brought by concentration in the lending market is exactly captured by the term in red in (23).

Summarizing: (23) shows how concentration increases the equilibrium DeFi rate y_n^{LS}, and identifies the sources of this increase. Terms in blue come from concentration in the staking market (what obtains when there is no negative return impact in the lending market); the term in red comes from concentration in the lending market (what obtains when there is no negative return impact in the staking market).

A similar decomposition can be done in the case of direct staking. However, there are two key differences between LS and DS:

Proposition 2. *(i) Under DS, concentration in the staking market does not impact borrowing*

[11] Details on the solutions of the comparison models are in Appendix A.6.

rates; under LS, it does: we always have

$$\Delta y_{s,n}^{DS} = 0, \tag{27}$$

$$\Delta y_{s,n}^{LS} > 0. \tag{28}$$

(ii) Under DS, concentration has a linear impact on borrowing rates; under LS, it has a superlinear impact: we always have

$$y_n^{DS} - \underline{y}^{DS} = \Delta y_{s,n}^{DS} + \Delta y_{\ell,n}^{DS}, \tag{29}$$

$$y_n^{LS} - \underline{y}^{LS} > \Delta y_{s,n}^{LS} + \Delta y_{\ell,n}^{LS}. \tag{30}$$

These results are important because they highlight that the "virtues of compounding" associated with liquid staking come with an undesirable byproduct, namely that compounding investments "imports" price impact from other markets and makes price impacts also compound. Specifically, the intuition for both points of Proposition 2 is the following.

For (i): under DS, the staking and lending markets are separated. Hence, the investors' entry condition on the staking market only impacts the equilibrium level of staking S (recall (24) and that the return on staking is R/S), not the required rate y on the lending market—which thus remains the same as in the case where no market is concentrated. By contrast, under LS, the blockchain-DeFi return results from the compounding of returns in both markets, and the upwards adjustment to the blockchain-DeFi return due to concentration mandated by (24) can only come from an increase in y (indeed, in equilibrium $S = \Omega$ which pins down the return on the staking market). By linking DeFi investment profitability to a market that was initially unrelated,[12] liquid staking "imports" the concentration and price impact issues from this market. Direct staking is not subject to this limitation.

For (ii): (29) adds an information relative to (27): concentration in the staking market

[12]We leave for future research the extension where the two frictions we consider—large agents and endogenous security—are combined, in which case staking market concentration may impact the DeFi lending market even under DS, through its impact on the equilibrium staking level and thus on blockchain security.

only does not distort borrowing rates under DS, but further, it does not *interact* with concentration in the lending market. That is, the price impact on y is the same whether there are concentration in both markets or in the lending market only. By contrast, (30) evidences an interaction between the price impacts in both markets, leading them to compound.

We also remark that the price impacts $\Delta y_{\ell,n}^{DS}$, $\Delta y_{\ell,n}^{LS}$ induced by concentration in the lending market only are of the same order of magnitude, a proportional factor $1/n$: from (26),

$$y_{\ell,n}^{LS} \approx \left(\frac{r}{1-\alpha} - 1\right)\left(1 + \frac{1}{n}\right) = \underline{y}^{LS}\left(1 + \frac{1}{n}\right), \text{ i.e. } \Delta y_{\ell,n}^{LS} \approx \frac{\underline{y}^{LS}}{n}, \tag{31}$$

and a similar equation holds for DS: see (64). Intuitively, an investor's price impact in the lending market is proportional to its contribution to the lending supply, which in a symmetric equilibrium is $1/n$, whether staking is liquid or direct.

Taken together, our results indicate that concentration in the DeFi outlet similarly impact borrowing rates under LS and DS; but then, LS also imports the price impact from the staking market and compounds it. This explains why, when concentration is strong, LS produces higher borrowing rates, which are detrimental to the end users of DeFi.

Having understood how LS can harm DeFi users, we now use our welfare metrics to include the other agents and thus assess the relative efficiency of both staking types. We start by using our closed-form expressions of welfare (Lemma 2), evaluated at the respective efficient rates y_n^{DS} and y_n^{LS}, to obtain the graphical representation of Figure 4.

For large n, liquid staking dominates direct staking, consistent with Proposition 1: indeed, the large agents friction vanishes and thus, as long as the delegation costs α are not too large, the capital efficiency channel makes liquid staking desirable. For small n, however, we see that direct staking is preferable. The main intuition comes from the investors' entry condition studied above. When n is small, the large agents friction is particularly impactful and increases investors' required rates of returns, and thus also borrowers' costs. This effect is magnified when staking is liquid due to the compounding of the price impacts, as formalized in Proposition 2.

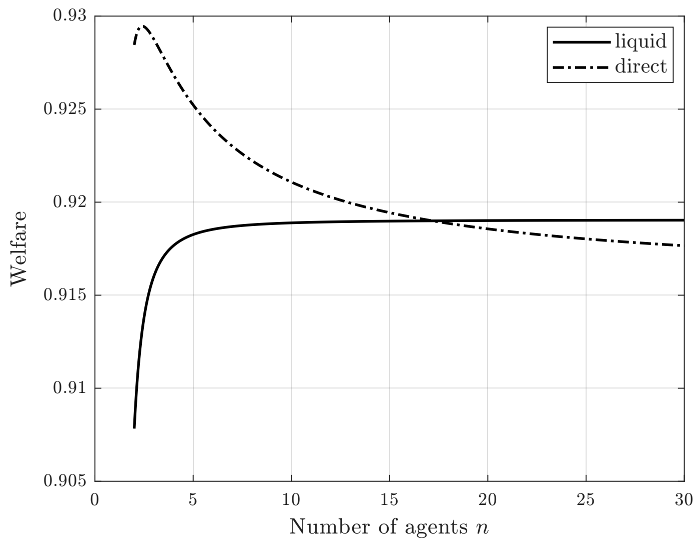

Figure 4: Welfares under liquid staking $W_n^{LS}(y_n^{LS})$ and under direct staking $W_n^{DS}(y_n^{DS})$ as a function of the number of investors n.

The Figure illustrates a single-crossing property: at the optimally adjusted DeFi rates, direct staking is more efficient for small n and liquid staking is more efficient for large n.

Thus, our model points at the fact that concentration impacts the relative efficiency of liquid versus direct staking. Unless the delegation costs α are so high or so low that they make one regime always superior to the other, this impact makes direct staking better for small n and worse for large n.

When α is large, direct staking is preferable even as the large agents friction vanishes, as seen in the benchmark model of Section 3 (where we derived and discussed the condition $\alpha \geq 1 - \frac{1}{\rho}$, under which LS is no longer preferable to DS). Given our earlier intuition that direct staking is even better when agents are large, in this case it is natural to expect that direct staking dominates liquid staking regardless of n.

When α is small, the relative efficiency of liquid staking only depends on the comparison between the capital efficiency channel and the compounding effect, as delegation costs no longer play a role. Absent these costs, there is no reason a priori that for all parameters, the compounding effect eventually dominates as n becomes small. This explains why we do not expect the single-crossing property for welfares to hold when α is too small.

Building on these intuitions, our next result establishes the single-crossing property formally under the condition that α is neither too small or too large:

Proposition 3. *For intermediate α ($\alpha_1^* < \alpha < \alpha_2^*$ with α_i^* given in the proof), there exists n^* such that liquid staking is less efficient than direct staking for $n \leq n^*$, and more efficient for $n > n^*$.*

To prove this result, we use the condition $\alpha < 1 - \frac{1}{\rho}$ guaranteeing that liquid staking is preferable in the benchmark case ($n \to \infty$) and derive another condition for the opposite to hold at $n = 2$. We then exhibit two additional technical conditions, bounding α above and below, under which the ratio of DS to LS welfares is decreasing in n. This is the simplest way to guarantee the single-crossing property.

Figure 5 illustrates that the single-crossing property is more general: it is obtained for parameters such that the welfare ratio is not monotonic, and yet shows that direct staking is preferable if and only if n is smaller than a threshold.

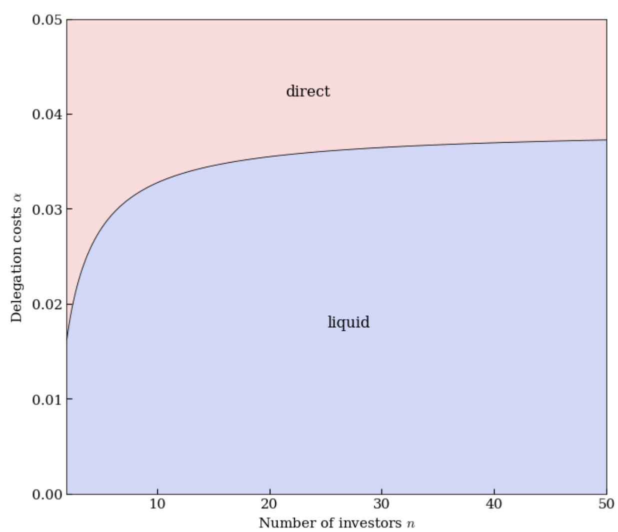

Figure 5: Better staking type as a function of the number of investors n and the liquid staking delegation cost α

In conclusion, we have shown that the large agents friction can make the introduction of liquid staking backfire. Indeed, there are configurations such that the LS delegation costs are sufficiently small that LS appears desirable if one assumes concentration away, but is in fact detrimental. The next section considers another friction which has the potential to render LS undesirable if not well designed.

5 Friction 2: Endogenous Security

So far, we assumed that the blockchain could not be attacked and was forever in operation (conditional on investors entering in the first place). In reality, the security of a Proof-of-Stake blockchain is a non-decreasing function of the market value of its staking pool, which in the LS model is given by $PS^{liq} = P\Omega$. To take this into account in a tractable manner, we assume that each period, the blockchain is attacked with probability $1-g$, with $g = \min\{\lambda PS^{liq}; 1\}$ and $\lambda > 0$. The potential attack takes place after investors made their allocation and before DeFi users decide how much to borrow. We assume that following an attack, the blockchain becomes unusable and the token price drops to zero. P_{t+1} now denotes next period's price *conditional on no attack*, and if the price is P now, the expected price next period is $\frac{gP}{\rho}$. Investors take into account the fact that the tokens they purchase in the CM at time t may be worthless in the next CM, and thus the LS blockchain equilibrium system is now:

$$g = \min\{\lambda P\Omega; 1\} \tag{L32}$$

$$r = (1+uy)(1-\alpha)g \tag{L33}$$

$$D = \frac{1}{1-\alpha}\frac{(1+y)^{-k}}{P} \tag{L34}$$

Similarly, under DS, the blockchain equilibrium system is now:

$$
\begin{aligned}
g &= \min\{\lambda PS; 1\} && \text{(D35)}\\
uy &= \frac{R}{S} && \text{(D36)}\\
r &= \frac{g}{\rho}(1+uy) && \text{(D37)}\\
D &= \frac{\rho}{P}(1+y)^{-k}. && \text{(D38)}
\end{aligned}
$$

As in the frictionless model, only DeFi users have their utility impacted by the design of the blockchain environment, and their utility is (i) a decreasing function of y when they can borrow (ii) minimal when they cannot borrow. The difference introduced by the endogenous security friction is that the equilibrium can be blockchain (i.e. investors enter initially) but the blockchain is later attacked. In this case, initial users borrow, but future users may not be able to do so. The probability that future users can access the lending pool is an increasing function of g (specifically, the blockchain has been attacked at time $\bar{t}$ or before with probability $1 - g^{t+1}$), and so (y, g) characterizes welfare.

The derivation of the equilibrium system, its solution, and the determination of the efficient DeFi rate under direct staking are done in Carré and Gabriel (2022). We now focus on solving the model under liquid staking. Denote $\varphi : \mathbb{R}_+ \to \mathbb{R}$ the function defined by $\varphi(x) = x(1+x)^{-k}$ and

$$
I = \underbrace{\left[\frac{r}{1-\alpha} - 1, +\infty\right[}_{I_1} \cap \underbrace{\varphi^{-1}\left(\left[\frac{r-1+\alpha}{\lambda}, +\infty\right[\right)}_{I_2}.
$$

By studying the function φ, it is easy to check that I_2, and thus I, are intervals.

Lemma 3 (equilibria under LS and endogenous security)**.** *When $y \in \mathbb{R}_+ \setminus I$, the equilibrium is no-blockchain and welfare is minimal. When $y \in I$:*

- *if y is interior to I, a unique secure and a unique risky equilibria with liquid staking co-exist; welfare is larger in the former.*

- *at* $y^{LS}_{\min} \equiv \inf I$*, there exists a unique equilibrium with liquid staking, and it is secure (and non-rationed).*

Lemma 3 says that the DeFi rate y must belong to $I = I_1 \cap I_2$ for a blockchain equilibrium to exist. The condition $y \in I_1$ simply states that $y \geq \frac{r}{1-\alpha} - 1 = \underline{y}^{LS}$, the minimal viable rate in the benchmark case, and was thus already encountered in Section 3. The condition $y \in I_2$ further restricts the set of viable DeFi rates: it guarantees that there exists consistent expectations about the blockchain's security level.

Once again, the efficient rate is the smallest one compatible with equilibrium:

Corollary 1 (optimal DeFi rate)**.** *The welfare-maximizing DeFi rate under liquid staking is* y^{LS}_{min}.

We can now state the main result of this section. It evidences that, in contrast with the one-sided results of Proposition 1, liquid staking can be detrimental when security is endogenous - *even for small or zero delegation costs.* If the "capital efficiency" intuition could be taken at face value, then in the absence of mechanical LS costs, one would conclude that replacing DS by LS in a given DeFi environment (here, fixing the DeFi borrowing rate y) must be beneficial. We prove this is not necessarily the case. We show, however, that a proper adjustment of the DeFi outlet, joint with the introduction of LS, does increase welfare unless delegation costs are prohibitive:

Proposition 4 (liquid staking can backfire)**.** *(i) At fixed* y*, even for small or zero* α*, welfare under liquid staking can be strictly smaller than under direct staking.*

(ii) However, liquid staking strictly welfare-dominates direct staking when y *is adjusted optimally (provided that* α *is not too large).*

Figure 6, which obtains for $\alpha = 0$, illustrates point (i) of Proposition 4. On a range of DeFi borrowing rates, blockchain security in the risky equilibrium is worse under liquid staking. In turn, this can imply a lower welfare when staking is liquid instead of direct. This result means that the "capital efficiency" intuition needs to be revisited. In fact, as we now

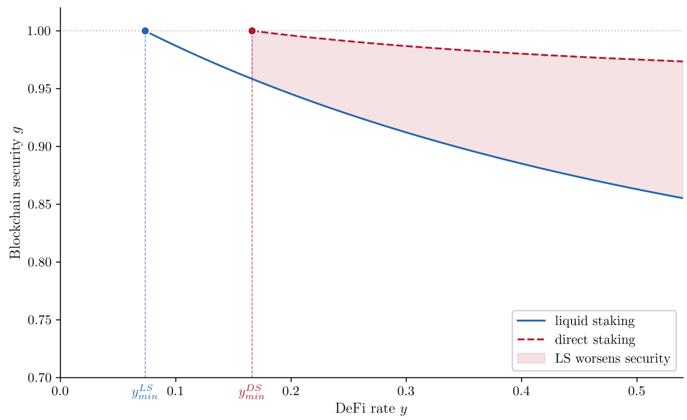

Figure 6: Security in the risky equilibrium

explain, the capital efficiency logic is meaningful, but can backfire under the endogenous security friction. For the purpose of isolating intuition, one can think of the investors' entry condition as a condition of the form

$$g\kappa Y \geq r, \tag{39}$$

where r is their outside option, Y is the gross return earned on DeFi, g is the endogenous probability of blockchain survival, and κ is a factor linked to the particular design considered: under DS, $\kappa = \frac{1}{\rho}$ i.e. investors are exposed to token inflation; under LS, $\kappa = 1 - \alpha$, i.e. investors lose the delegation cost.

If $\kappa Y > r$, then anticipating $g = 1$ or $g < 1$ can a priori be compatible with equilibrium. In words: is the expected return great because the investment is great, or simply because it is risky? For instance, if $\kappa Y = 110\%$ and $r = 103\%$, it may be consistent to anticipate that the extra 7% is a compensation for attack risk. Hence the co-existence of a safe and a risky equilibrium. But if $\kappa Y = r$, the only expectation consistent with equilibrium is $g = 1$. At the optimal DS rate, which enforces that the only equilibrium features $g = 1$, we thus have $\kappa Y^* = r$. *Precisely because liquid staking increases capital efficiency,* using LS instead of DS corresponds to increasing κ (as long as delegation costs are not too high). Indeed, under LS investors no longer undergo a loss due to token inflation ρ. But this very fact implies that under LS, at the same DeFi rate, $\kappa Y^* > r$, making $g < 1$ again compatible with the entry condition (39): under an "intrisically" better mechanism (i.e. that produces a priori more returns for investors), investors are ready to enter even if there is some risk; under a "worse" mechanism, investors are only willing to enter if there is no risk, as they would not accept any further reduction in their expected return. It is in this sense that liquid staking backfires: in the above example where the DeFi rate was fixed, moving from DS to LS reintroduces a level of security $g < 1$ consistent with equilibrium.

It is important to note that the previous reasoning indicates that the *maximal* risk level compatible with investors entry decreases as the interest rate decreases. The fact that this upper bound decreases leaves open the question of how the security level that actually realizes in the risky equilibrium comoves with the DeFi interest rate. We answer this question in Appendix A.11.

Point (ii) of the Proposition, however, indicates that unless α is too large, LS does increase welfare if the DeFi borrowing rate is optimally adjusted. The intuition again follows from the explanation above: at the optimal rate $y_{\min}^{LS}$ exhibited in Lemma 3, the risky equilibrium ($g < 1$) disappears. Indeed, with the notation used above, we have $\kappa Y_{\min}^{LS} = r$ and thus $g\kappa Y_{\min}^{LS} \geq r$ is only possible if $g = 1$. Thus, when the DeFi rate is optimally adjusted, the equilibrium is safe both under LS and DS, but borrowers face lower costs under LS, making LS more efficient. This is depicted in Figures 7 and 8. The former illustrates that, for $\alpha = 0$, once DeFi rates are optimally adjusted, LS does provide larger welfare than DS. The latter exhibits the threshold α^* below which the delegation cost must lie for LS to be more efficient than DS.[13]

[13]The "kinks" in Figures 7 and 8 correspond to a change of regime where utilization under direct staking switches from being equal to 1 to being strictly below 1. The difference between these two regimes is already explored in Carré and Gabriel (2022), so we do not discuss it further here.

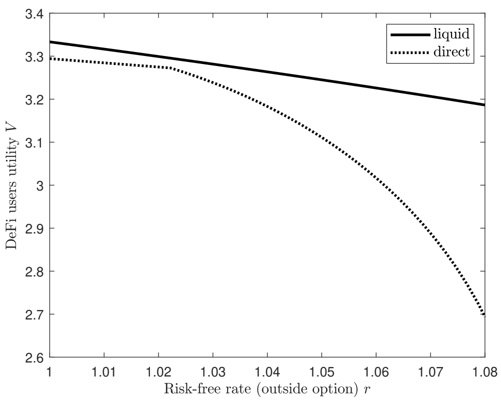

Figure 7: Welfare comparison across staking types

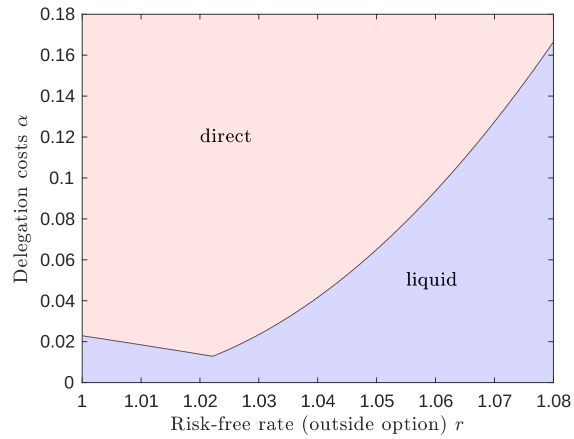

Figure 8: Better staking type

Conclusion

We have built a tractable overlapping generations, general equilibrium model to compare the efficiency of direct versus liquid staking. In its benchmark version without friction, the model confirms the "capital efficiency" intuition: liquid staking is preferable, indeed because it allows to compound investments. However, this can backfire when a realistic friction is introduced. When investors are large, liquid staking "imports" price impact issues from the staking market to DeFi, and makes price impacts compound. As a result, investors require larger rates of return on DeFi, which harms DeFi users. When the security of the blockchain is endogenous to the market value of its staking pool, the very fact that the liquid staking mechanism is designed to be more efficient reopens the possibility of a risky equilibrium, unless the design of the DeFi outlet is properly adjusted when liquid staking is introduced. The reason is that investors may rationally coordinate on the belief that the DeFi rate is "abnormally large" because it contains compensation for attack risk.

Several avenues are interesting for future research. In the context of our model and as mentioned earlier, one could study the interaction of the two frictions we considered. Studying further the "companion innovation" of liquid staking, namely restaking,[14] also appears important.

[14] See e.g. Chitra and Pai (2024).

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A Additional Proofs

A.1 Benchmark: Equilibrium

The benchmark equilibrium characterization is straightforward and is given in the main text. However, there we focused on non-rationed equilibria; hence we still need to treat the case of rationed equilibria.

Liquid staking. In a rationed equilibrium, $u = 1$ and $D = \Omega$. (L1) indicates that a safe rationed equilibrium can only exist if the DeFi borrowing rate is

$$y = \frac{r}{1-\alpha} - 1. \tag{40}$$

The DeFi users' equation (L2), which characterized the desired level of borrowing D^*, transforms into $D^* > D = \Omega$ i.e.

$$P < \overline{P} \equiv \frac{(1+y)^{-k}}{(1-\alpha)\Omega}. \tag{41}$$

$\overline{P}$ is the safe, non-rationed, equilibrium price prevailing at the DeFi borrowing rate (40). The Walrasian auctioneer has the possibility to clear their centralized market by selecting $\overline{P}$ and thus rule out rationed equilibria that would appear for $P < \overline{P}$. It is optimal to do so as it increases DeFi users' utility. Indeed, using equations (4) and (5), this utility writes

$$V = v\left(Q_t\right) - e_{\underline{t+1}} = \frac{1}{1-\frac{1}{k}} x^{1-\frac{1}{k}} - x\left(1+y\right), \tag{42}$$

where $x = PD(1-\alpha)$. That DeFi users are constrained to borrow only $D = \Omega$ in a rationed equilibrium means that they would be better off if they could borrow more than this quantity, i.e. the RHS of (42) has positive derivative at the equilibrium value of x. Thus, increasing P, which increases x, increases V and hence is optimal as long as we are in a rationed equilibrium.

Direct staking. The logic above still applies: the Walrasian auctioneer of the CM optimally eliminates rationed equilibria. The proof is similar and corresponds to the particular case $g = 1$ of the exogenous security benchmark analyzed in Carré and Gabriel (2022).

A.2 Benchmark: Proposition 1

Recall (8) and (10): $\underline{y}^{LS} = \frac{r}{1-\alpha} - 1$ and $\underline{y}^{DS} = \rho r - 1$. Hence, $\underline{y}^{LS} < \underline{y}^{DS}$ if and only if $\frac{1}{1-\alpha} < \rho$, i.e. $\alpha < 1 - \frac{1}{\rho}$, which is the condition given in the Proposition. When it is met, there are three possible situations. a) When $y < \underline{y}^{LS}$, equilibrium is no-blockchain both under LS and DS, and welfares are thus identical (and minimal). b) When $y \geq \underline{y}^{DS}$, equilibrium is blockchain both under LS and DS, and since, in the frictionless model, welfare is purely driven by y, welfares are identical. c) When $\underline{y}^{LS} \leq y < \underline{y}^{DS}$, equilibrium is blockchain under LS and no-blockchain under DS, thus welfare under LS is strictly larger. This proves (i). (ii) is then immediate, as the optimal rate under LS is $\underline{y}^{LS}$, which creates a blockchain equilibrium featuring a strictly smaller y than the lowest achievable y under DS, namely $\underline{y}^{DS}$.

A.3 Friction 1: Symmetric equilibrium conditions

As announced in section 4, we justify equilibrium equations (L11) (liquid staking equilibrium) and (D13) (direct staking equilibrium).

Liquid staking. Each investor solves:

$$\max_{s} r\left(\frac{e}{n} - Ps\right) + \frac{P}{\rho}s(1-\alpha)\left(1 + \frac{R}{s+S_-}\right)\left(1 + \frac{D}{s+S_-}y\right), \tag{43}$$

where $\frac{e}{n}$ is their initial endowment, s is the quantity of tokens they buy and liquid stake, and S_- is the total quantity staked by others. The first-order condition yields

$$\begin{aligned} r &= \frac{1}{\rho}(1-\alpha)\left(1 + \frac{R}{s+S_-}\right)\left(1 + \frac{D}{s+S_-}y\right) \\ &+ \frac{1}{\rho}s(1-\alpha)\left(-\frac{R}{(s+S_-)^2}\right)\left(1 + \frac{D}{s+S_-}y\right) \\ &+ \frac{1}{\rho}s(1-\alpha)\left(1 + \frac{R}{s+S_-}\right)\left(-\frac{D}{(s+S_-)^2}y\right). \end{aligned}$$

In a symmetric equilibrium, $S_- = (n-1)s$ and $S_- + s = \Omega$. The above equality thus simplifies to

$$\frac{r}{1-\alpha} = \frac{1}{\rho}\left[(1+uy)\left[\frac{n-1}{n}\rho + \frac{1}{n}\right]\right] - \frac{1}{n}uy, \tag{44}$$

which is indeed (L11).

Direct staking. Investors now solve:

$$\max_{s,\ell} r\left(\frac{e}{n} - Ps - P\ell\right) + \frac{P}{\rho}\left[s\left(1 + \frac{1}{s+S_-}R\right) + \ell\left[1 + \frac{D}{\ell+L_-}y\right]\right], \tag{45}$$

where s and ℓ are their token allocations to staking and lending, and S_- and L_- are the total quantities staked and lent by others. The first-order conditions are

$$r = \frac{1}{\rho}\left[1 + \frac{S_-}{(s+S_-)^2}R\right], \qquad r = \frac{1}{\rho}\left[1 + \frac{L_-}{(\ell+L_-)^2}Dy\right].$$

In a symmetric equilibrium, $S_- = (n-1)s$ and $L_- = (n-1)\ell$. $S = ns$ is the total amount of staking and $L = n\ell$ is the total amount of lending. The first-order conditions then simplify to

$$r = \frac{1}{\rho}\left[1 + \frac{(n-1)}{n}\frac{R}{S}\right], \qquad r = \frac{1}{\rho}\left[1 + \frac{(n-1)}{n}uy\right].$$

This justifies (D13), and also implies the other equilibrium condition (D14), i.e. $uy = \frac{R}{S}$.

A.4 Friction 1: Lemma 1

Liquid staking. Recall Equation (L11):

$$\frac{r}{1-\alpha} = (1+uy)c_n - \frac{1}{n}uy,$$

where $c_n = \frac{1}{\rho}\left[\frac{n-1}{n}\rho + \frac{1}{n}\right]$. This gives the unique candidate value of u:

$$u = \frac{\frac{r}{1-\alpha} - c_n}{y(c_n - \frac{1}{n})}.$$

Note that this is always positive. Hence, for this to be part of an equilibrium, we only need to ensure that $u \leq 1$, i.e., that the rate y is not too low: $y_n^{LS} := \frac{\frac{r}{1-\alpha} - c_n}{(c_n - \frac{1}{n})} \leq y$. By definition of the utilization, $D = u\Omega$ and thus, by Equation (L12), the price is given by Equation (18).

Direct staking. Equations (D13) and (D14) imply that $r = \frac{1}{\rho}\left[1 + \frac{(n-1)}{n}uy\right]$, hence $u = \frac{n}{n-1}\frac{(r\rho-1)}{y}$. This is always positive, however, we must ensure that $u \leq 1$. This is equivalent to

$$y_n^{DS} := \frac{n}{n-1}(r\rho - 1) \leq y.$$

Then, by Equation (D13), $S = \frac{n-1}{n}\frac{\rho-1}{r\rho-1}\Omega$, and by Equation (D16), $L = \left(1 - \frac{n-1}{n}\frac{\rho-1}{r\rho-1}\right)\Omega = \frac{n\rho(r-1)+\rho-1}{n(r\rho-1)}\Omega$. Demand is then $D = uL = \frac{1}{n-1}\frac{n\rho(r-1)+\rho-1}{y}\Omega$. Hence, by Equation (D15), the price is given by $P = \frac{\rho}{D}(1+y)^{-k} = \frac{\rho(n-1)y}{\Omega(n\rho(r-1)+\rho-1)}(1+y)^{-k}$.

Once again, we need to conclude the equilibrium characterization by the study of rationed equilibria, showing that they can be eliminated. In this case, $u = 1$, and, the only change to equilibrium equations is that the equality giving the value of the DeFi users' level of borrowing D^* (equation (L12)) is replaced by $D^* > \Omega$, which again gives the equivalent condition $P \leq \overline{P}$ (this holds for LS, see equation (41)—as explained in A.1 the case of DS is similar, it just changes the value of the upper bound $\overline{P}$ of token prices compatible with a rationed equilibrium.) $\overline{P}$ is the market-clearing price corresponding to the non-rationed equilibrium. By selecting $P = \overline{P}$, the Walrasian auctioneer of the CM eliminates rationed equilibria; however the question is whether this action is still optimal. The fact that DeFi users' utility is increasing in P across rationed equilibria still holds (no change to the proof of A.1), but now, under Friction 1, investors also have non-constant utility and thus impact welfare. To conclude, we prove that their utility is also increasing in P across rationed equilibria.

- *Liquid staking.* At equilibrium, $s = \frac{\Omega}{n}$ and thus the utility of an individual investor, given in (43), simplifies in equilibrium (rationed or not) as

$$r\left(\frac{e}{n} - P\frac{\Omega}{n}\right) + P\frac{\Omega}{n}(1-\alpha)(1+uy). \tag{46}$$

  Recall that $u = 1$ in a rationed equilibrium; hence the slope in P is proportional to

$$-r + (1-\alpha)(1+y) > (1-\alpha)(1+y) - (1-\alpha)(1+y)\left(\frac{n-1}{n} + \frac{1}{n\rho}\right) > 0,$$

where the first inequality follows from (L11) and the second one follows from $\rho > 1$.

- *Direct staking.* Using $s + \ell = \frac{\Omega}{n}$ and $uy = \frac{R}{S}$, the utility of an individual investor, given in (45), simplifies in equilibrium (rationed or not) as

$$r\left(\frac{e}{n} - P\frac{\Omega}{n}\right) + \frac{P}{\rho}\frac{\Omega}{n}\left(1 + \frac{R}{S}\right) = r\frac{e}{n} + P\frac{\Omega}{n}\left[\frac{1}{\rho}\left(1 + \frac{R}{S}\right) - r\right] = r\frac{e}{n} + P\frac{\Omega}{n^2}\frac{1}{\rho}\frac{R}{S}, \tag{47}$$

  using (D13).

Thus, in both cases, the utility of investors increases in P across rationed equilibria, and so does welfare, which concludes the proof.

A.5 Friction 1: Lemma 2

Liquid staking. At equilibrium, the utility of an individual investor is given by (46). Summing and adding the utility of DeFi users—which is still given by (6)—we find that welfare under liquid staking is

$$re + P\Omega\left((1-\alpha)(1+uy) - r\right) + \frac{1}{k-1}(1+y)^{1-k}.$$

Using (17) and (18), we obtain

$$\begin{aligned} W_{n,\alpha}^{LS}(y) &= re + \left[\left((1-\alpha-r)\frac{c_n - \frac{1}{n}}{r - c_n(1-\alpha)} + 1 + \frac{1}{k-1}\right)y + \frac{1}{k-1}\right](1+y)^{-k} && (48)\\ &= re + \left[\left(\frac{(1-\alpha-r)}{(1-\alpha)}\frac{1}{y_n^{LS}} + 1 + \frac{1}{k-1}\right)y + \frac{1}{k-1}\right](1+y)^{-k}. && (49)\end{aligned}$$

Direct staking. At equilibrium, the utility of an individual investor is given by (47). Summing and adding the utility of DeFi users (6), we find that welfare under direct staking is

$$re + P\frac{\Omega}{n}\frac{1}{\rho}\frac{R}{S} + \frac{1}{k-1}(1+y)^{1-k}.$$

Using $\frac{R}{S} = uy$ and the values of u and P, we obtain

$$\begin{aligned} W_{n}^{DS}(y) &= re + \left[\left(\frac{r\rho - 1}{n\rho(r-1) + \rho - 1} + \frac{1}{k-1}\right)y + \frac{1}{k-1}\right](1+y)^{-k} && (50)\\ &= re + \left[\left(\frac{y_n^{DS}}{(y_n^{DS} + 1 - \rho)n} + \frac{1}{k-1}\right)y + \frac{1}{k-1}\right](1+y)^{-k}. && (51)\end{aligned}$$

We now show that both welfare functions are decreasing in y. We will repeatedly use the following observation: a function of the form $y \mapsto (ay+b)(1+y)^{-k}$ is decreasing for large y if $a(1-k) < 0$, and in that case the threshold after which it decreases is $y^* := \frac{kb-a}{a(1-k)}$.

Liquid staking. The function $y \mapsto W_{n,\alpha}^{LS}(y)$ is of this form, hence to show that it decreases for $y \geq y^*$, we need to prove that $\frac{(1-\alpha-r)}{(1-\alpha)}\frac{1-k}{y_n^{LS}} - k < 0$. When $k \leq 1$, this is obviously true since the L.H.S. is negative. When $k > 1$, this is equivalent to $\frac{r}{1-\alpha} - 1 < y_n^{LS}\frac{k}{k-1}$, which holds since $y_n^{LS} \geq \frac{r}{1-\alpha} - 1$ and $\frac{k}{k-1} > 1$. Hence $W_{n,\alpha}^{LS}$ is decreasing for $y \geq y^*$, so it will indeed be decreasing over the range of admissible values of y as soon as

$$y_n^{LS} > y^* = \frac{r+\alpha-1}{1-\alpha}\frac{1}{y_n^{LS}}\left(\frac{1}{\frac{1-\alpha-r}{1-\alpha}\frac{1-k}{y_n^{LS}}-k}\right).$$

The denominator in the parenthesis is negative for $k<1$ (as $1-\alpha-r<0$). But it is also negative for $k\geq 1$, given the inequality $y_n^{LS}\geq \frac{r}{1-\alpha}-1$. Hence $y^*<0$ and $y_n^{LS}>y^*$, showing that y_n^{LS} is the optimal rate.

Direct staking. We use the same approach, since $y\mapsto W_{n,\alpha}^{DS}(y)$ also has the form given in our preliminary observation. To show that it decreases after y^*, we need

$$\left(\frac{y_n^{DS}}{(y_n^{DS}+1-\rho)n}+\frac{1}{k-1}\right)(1-k)<0. \tag{52}$$

But $y_n^{DS}=\frac{n}{n-1}\left(r\rho-1\right)\geq\rho-1$, so

$$(y_n^{DS}+1-\rho)n\geq 0. \tag{53}$$

Hence when $k\geq 1$ the L.H.S. is clearly negative. When $k<1$, we need to show that $\frac{y_n^{DS}}{(y_n^{DS}+1-\rho)n}+\frac{1}{k-1}<0$ or equivalently $y_n^{DS}>\frac{(\rho-1)n}{k+n-1}$, which is true since $y_n^{DS}=\frac{(r\rho-1)n}{n-1}$. Finally, we need to verify that

$$y_n^{DS}>y^*=\left(y_n^{DS}(n-1)+(1-\rho)n\right)\frac{1}{(y_n^{DS}+1-\rho)n}\frac{1}{\left(\frac{y_n^{DS}}{(y_n^{DS}+1-\rho)n}+\frac{1}{k-1}\right)(1-k)}.$$

Since $y_n^{DS}=\frac{n}{n-1}\left(r\rho-1\right)$, we have $y_n^{DS}(n-1)+(1-\rho)n=n\rho\left(r-1\right)>0$ hence, from Equations (52) and (53) we conclude that y^* is negative. So $y_n^{DS}>y^*$, showing that y_n^{DS} is the optimal rate.

A.6 Friction 1: Comparison models and Proposition 2

Liquid staking, fixed returns in the lending market. The only difference with the main model is that investors now solve:

$$\max_s r\left(\frac{e}{n}-Ps\right)+\frac{P}{\rho}s(1-\alpha)\left(1+\frac{R}{s+S_-}\right)(1+y) \tag{54}$$

(set utilization at one in (43).) F.O.C.:

$$r=\frac{1+y}{\rho}(1-\alpha)\left(1+R\frac{S_-}{(s+S_-)^2}\right). \tag{55}$$

Using $R=\Omega(\rho-1)$ and the symmetric equilibrium conditions $s=\frac{S_-}{n-1}=\frac{\Omega}{n}$ gives

$$r=\frac{1+y}{\rho}(1-\alpha)\left(1+(\rho-1)\frac{n-1}{n}\right), \tag{56}$$

and isolating y gives the formula of the text for $y_{s,n}^{LS}$.

Liquid staking, fixed returns in the staking market. The only difference with the main model is that investors now solve:

$$\max_s r\left(\frac{e}{n} - Ps\right) + \frac{P}{\rho}s(1-\alpha)(1+\beta)\left(1 + \frac{D}{s+S_-}y\right) \tag{57}$$

(set the per-unit staking reward at some fixed β in (43).) F.O.C.:

$$r = \frac{1+\beta}{\rho}(1-\alpha)\left(1 + Dy\frac{S_-}{(s+S_-)^2}\right). \tag{58}$$

Using the fact that the relevant benchmark is $\beta = \frac{R}{\Omega} = \rho - 1$, the symmetric equilibrium conditions $s = \frac{S_-}{n-1} = \frac{\Omega}{n}$, we find

$$r = (1-\alpha)\left(1 + uy\frac{n-1}{n}\right). \tag{59}$$

The optimal y is obtained at $u = 1$: isolating this y gives the formula of the text for $y^{LS}_{\ell,n}$.

Direct staking, fixed returns in the lending market. The only difference with the main model is that investors now solve:

$$\max_{s,\ell} r\left(\frac{e}{n} - Ps - P\ell\right) + \frac{P}{\rho}\left[s\left(1 + \frac{1}{s+S_-}R\right) + \ell\,(1+y)\right], \tag{60}$$

(set utilization at one in (45).) The F.O.C. in ℓ implies that $r = \frac{1+y}{\rho}$, which means that $y^{DS}_{s,n} = \underline{y}^{DS}$.

Direct staking, fixed returns in the staking market. The only difference with the main model is that investors now solve:

$$\max_{s,\ell} r\left(\frac{e}{n} - Ps - P\ell\right) + \frac{P}{\rho}\left[s\,(1+\beta) + \ell\left(1 + \frac{D}{\ell+L_-}y\right)\right] \tag{61}$$

(set the per-unit staking reward at some fixed β in (45).) The F.O.C. in ℓ gives

$$r = \frac{1}{\rho}\left(1 + Dy\frac{L_-}{(\ell+L_-)^2}\right). \tag{62}$$

Using the fact the symmetric equilibrium condition $\ell = \frac{L_-}{n-1} = \frac{L}{n}$, we find

$$r = \frac{1}{\rho}\left(1 + uy\frac{n-1}{n}\right). \tag{63}$$

The optimal y is obtained at $u = 1$: isolating this y gives $y^{DS}_{\ell,n} = \frac{n}{n-1}(\rho r - 1)$, which is also equal to y^{DS}_n. We can also rewrite

$$y^{DS}_n = \frac{\overbrace{\rho r - 1}^{\text{frictionless } \underline{y}^{DS}}}{1 - \frac{1}{n}}, \tag{64}$$

mirroring (23). This confirms that, as discussed in the main text, under DS concentration in the staking market has no impact on borrowing rates (the terms in blue in (23) are no longer present), and that concentration in the lending market has a similar proportional impact (the term in red is

$1/n$ both under LS and DS).

Proof of Proposition 2. (i) is already proven, since we have seen above that $y_{s,n}^{DS} = \underline{y}^{DS}$, and $\Delta y_{s,n}^{LS} > 0$ is immediate from the expression of $y_{s,n}^{LS}$ given in the main text.

(ii) The verification of (29) is immediate. We thus now prove (30), which is equivalent to

$$y_n^{LS} + \underline{y}^{LS} > y_{s,n}^{LS} + y_{\ell,n}^{LS}. \tag{65}$$

Let $b = \frac{\rho-1}{n\rho}$ and $\underline{y} = \underline{y}^{LS}$ to simplify the expressions of the various rates. The inequality rewrites

$$\frac{\underline{y}+b}{1-b-\frac{1}{n}} + \underline{y} > \frac{\underline{y}+b}{1-b} + \frac{\underline{y}}{1-\frac{1}{n}}, \tag{66}$$

i.e.

$$(\underline{y}+b)f\left(b+\frac{1}{n}\right) + \underline{y}f(0) > (\underline{y}+b)f(b) + \underline{y}f\left(\frac{1}{n}\right), \tag{67}$$

where $f(x) = \frac{1}{1-x}$ for $0 \leq x < 1$. This is equivalent to

$$(\underline{y}+b)\left(f\left(b+\frac{1}{n}\right) - f(b)\right) > \underline{y}\left(f\left(\frac{1}{n}\right) - f(0)\right), \tag{68}$$

which is true because $b > 0$ and f is convex, so that its increment over $[b; b+\frac{1}{n}]$ is larger than its increment over $[0; \frac{1}{n}]$.

A.7 Friction 1: Proposition 3

For conciseness, we focus on the case $k > 1$. The proof is decomposed as follows. We begin by rewriting the expressions of welfare in a convenient manner. Then we find conditions (i) under which LS dominates for n large, (ii) under which DS dominates at $n = 2$, (iii) under which the ratio of DS to LS welfare is decreasing. Taken together, these conditions imply the desired single-crossing property.

Convenient expressions of welfare. Let $q = \frac{r}{1-\alpha}$. We always consider welfare at the optimally adjusted DeFi rate, and thus omit this dependency for conciseness, e.g. W_n^{LS} denotes $W_n^{LS}(y_n^{LS})$.

For liquid staking, from Equation (49) (recall that we no longer report the constant re term), we have:

$$W_n^{LS} = \left[\frac{k}{k-1}(1+y_n^{LS}) - q\right](1+y_n^{LS})^{-k}, \tag{69}$$

where $1 + y_n^{LS} = \frac{nq-1}{n-(2-\frac{1}{\rho})}$. Note that since $y_n^{LS} \to q-1$ as $n \to \infty$, we get that

$$W_n^{LS} \underset{n\to\infty}{\longrightarrow} \frac{1}{k-1}q^{-k+1}. \tag{70}$$

For direct staking, from Equation (51), we have:

$$W_n^{DS} = \left[\left(\frac{y_n^{DS}}{(y_n^{DS}+1-\rho)n} + \frac{1}{k-1}\right)y_n^{DS} + \frac{1}{k-1}\right](1+y_n^{DS})^{-k},$$

where $1 + y_n^{DS} = \frac{nr\rho-1}{n-1}$. Since $y_n^{DS} \to r\rho - 1$ as $n \to \infty$, we get that

$$W_{DS}^{n} \underset{n\to\infty}{\longrightarrow} \frac{1}{k-1}(r\rho)^{-k+1}. \tag{71}$$

For $n \geq 2$, let

$$\Phi(n) = \frac{W_n^{DS}}{W_n^{LS}}.$$

Using the expression above, we calculate Φ as

$$\begin{aligned}
\Phi(n) &= \left(\frac{1+y_n^{LS}}{1+y_n^{DS}}\right)^k \left(\frac{n-(2-\frac{1}{\rho})}{n-1}\right) \left(\frac{nr\rho - 1 + (k-1)\frac{n(r\rho-1)^2}{n(r\rho-\rho)+\rho-1}}{nq - k + q\left(2-\frac{1}{\rho}\right)(k-1)}\right) \\
&= \left(\frac{n-1}{n-(2-\frac{1}{\rho})}\right)^{k-1} \left(\frac{nq-1}{nr\rho-1}\right)^k \frac{N(n)}{D(n)},
\end{aligned}$$

with

$$\begin{aligned}
N(n) &= nr\rho - 1 + (k-1)\frac{n(r\rho-1)^2}{n(r\rho-\rho)+\rho-1}, \\
D(n) &= nq - k + q\left(2-\frac{1}{\rho}\right)(k-1).
\end{aligned}$$

Condition for LS to be better when $n \to \infty$. Comparing the asymptotic expressions (70) and (71) and recalling $q = \frac{r}{1-\alpha}$ shows that as soon as $\rho r > q$, i.e.

$$\alpha < 1 - \frac{1}{\rho}, \tag{72}$$

welfare under LS is larger than under DS for n large.
Condition for DS to be better when $n = 2$. Verify the intuitive fact that LS welfare decreases in α (i.e. in q):

$$\frac{d}{dq}\ln(W_n^{LS}) = (1-k)\frac{(1-n)^2\frac{1}{\rho} + (2+n(n+k-3))(1-\frac{1}{\rho}) + n(q-1)\left(k(2-\frac{1}{\rho}) + n - (2-\frac{1}{\rho})\right)}{(k(q(2-\frac{1}{\rho})-1) + q(n-(2-\frac{1}{\rho})))(nq-1)} \tag{73}$$

$1-k<0$, and both the numerator and denominator are a sum of positive terms, since $n \geq 2$, $k > 1$ and $\rho > 1$. Hence W_n^{LS} decreases in q. In the particular case where $n = 2$, this result—combined with the fact that, from (69) and the expression of y_n^{LS} as a function of q, LS welfare goes to 0 as q goes to ∞—proves the existence of q_1^* such that $q > q_1^* \implies W_{n=2}^{DS} > W_{n=2}^{LS}$, i.e. $\Phi(2) > 1$. In terms of α, the condition is

$$\alpha > 1 - \frac{r}{q_1^*}. \tag{74}$$

Conditions for the welfare ratio to be decreasing in n. We extend the domain of definition of our functions to the real interval $[2, +\infty[$ and set to show that, under conditions that we explicit, Φ is decreasing over this interval.

$$
\begin{aligned}
\log\Phi(n) &= (k-1)\log\left(\frac{n-1}{n-(2-\frac{1}{\rho})}\right) + k\log\left(\frac{nq-1}{nr\rho-1}\right) + \log N(n) - \log D(n),\\
\frac{d}{dn}\log\Phi(n) &= (k-1)\left(\frac{1}{n-1} - \frac{1}{n-(2-\frac{1}{\rho})}\right) + k\left(\frac{q}{nq-1} - \frac{r\rho}{nr\rho-1}\right) + \underbrace{\frac{N'(n)}{N(n)} - \frac{D'(n)}{D(n)}}.
\end{aligned}
$$

We first find a condition under which the bracket is negative, which, given the positivity of N and D, boils down to showing that

$$
F(n) := N'(n)D(n) - N(n)D'(n) \le 0.
$$

F is decreasing because

$$
F'(n) = -2(k-1)(r-1)(\rho-1)(r\rho-1)^2 \frac{(2kq\rho - kq - k\rho + nq\rho - 2q\rho + q)}{(nr\rho - n\rho + \rho - 1)^3} \le 0.
$$

So we will indeed have $F(n) \le 0$ for $n \ge 2$ as soon as $F(2) \le 0$. This is equivalent to

$$
\frac{N'(2)}{N(2)} \le \frac{q}{q\left(2 + (k-1)\left(2 - \frac{1}{\rho}\right)\right) - k}.
$$

The R.H.S. is decreasing in q for $q \ge r$ (which holds by definition of q). Thus, there exists $q_2^* \in]0, +\infty]$ such that $F(2) \le 0$ for $q \le q_2^*$. In terms of α, the condition is

$$
\alpha \le 1 - \frac{r}{q_2^*}. \tag{75}
$$

Then:

$$
\begin{aligned}
\frac{d}{dn}\log\Phi(n) &\le (k-1)\left(\frac{1}{n-1} - \frac{1}{n-(2-\frac{1}{\rho})}\right) + k\left(\frac{q}{nq-1} - \frac{r\rho}{nr\rho-1}\right) &(76)\\
&\le -(k-1)\frac{1-\frac{1}{\rho}}{(n-1)(n-(2-\frac{1}{\rho}))} + k\frac{r\rho - q}{(nq-1)(nr\rho-1)}. &(77)
\end{aligned}
$$

Since $2 - \frac{1}{\rho} > 1$ the first term is smaller than $-(k-1)\frac{1-\frac{1}{\rho}}{(n-1)^2}$. Given that $q < r\rho$, and that, since $q, r, \rho > 1$, $nq - 1 \ge q(n-1)$ and $nr\rho - 1 \ge r\rho(n-1)$, the second term is smaller than $\frac{r\rho - q}{r\rho q(n-1)^2}$. Hence:

$$
\frac{d}{dn}\log\Phi(n) \le -\frac{1}{(n-1)^2}\underbrace{\left((k-1)(1-\frac{1}{\rho}) - k\frac{r\rho - q}{qr\rho}\right)}. \tag{78}
$$

Now observe that $\frac{r\rho - q}{qr\rho} = \frac{r\rho - \frac{r}{1-\alpha}}{qr\rho} = \frac{\rho(1-\alpha)-1}{q\rho(1-\alpha)} \le \frac{\rho(1-\alpha)-1}{\rho(1-\alpha)}$. This proves that the bracket in (78) is positive as soon as $(k-1)(1-\frac{1}{\rho}) - k\frac{\rho(1-\alpha)-1}{\rho(1-\alpha)} \ge 0$. This condition is equivalent to

$$
\alpha \ge 1 - \frac{k}{\rho + k - 1}. \tag{79}
$$

Conclusion. Taken together, conditions (72),(74), (75) and (79) are equivalent to

$$\alpha_1^* < \alpha < \alpha_2^*, \tag{80}$$

where

$$\begin{aligned}\alpha_1^* &:= \max\left\{1 - \frac{r}{q_1^*}; 1 - \frac{k}{\rho + k - 1}\right\},\\ \alpha_2^* &:= \min\left\{1 - \frac{r}{q_2^*}; 1 - \frac{1}{\rho}\right\}.\end{aligned}$$

When (80) holds, the ratio of welfares is above 1 at $n = 2$, below one for n large, and decreasing, which establishes the single-crossing property stated in the Proposition. As an example, the parameter set $k = 1.3, \alpha = 3\%, \rho = 1.04, r = 1.05$ satisfies (80).

A.8 Friction 2: Lemma 3

Four types of equilibria with liquid staking are a priori possible: an equilibrium can be safe or risky, and non-rationed or rationed. We analyze the conditions of existence of each of these types in turn, and then deduce the statement of the Lemma.

Safe non-rationed equilibria. From (L33) we obtain that utilization u must satisfy

$$u = \frac{1}{y}\left(\frac{r}{1-\alpha} - 1\right). \tag{81}$$

Since we must have $0 \leq u \leq 1$, we have the condition

$$r \leq (1-\alpha)(1+y), \tag{82}$$

which is equivalent to

$$y \in I_1 \equiv \left[\frac{r}{1-\alpha} - 1, +\infty\right[, \tag{83}$$

the interval introduced before the statement of the Lemma.

We also have $D = u\Omega$ and, from (L34),

$$P = \frac{(1+y)^{-k}}{(1-\alpha)D}. \tag{84}$$

In a safe equilibrium the inequality $\lambda P \Omega \geq 1$ holds, which leads to the second condition

$$r - 1 + \alpha \leq \lambda y \,(1+y)^{-k}, \tag{85}$$

which is equivalent to

$$y \in I_2 \equiv \varphi^{-1}\left(\left[\frac{r-1+\alpha}{\lambda}, +\infty\right[\right), \tag{86}$$

the interval introduced before the statement of the Lemma (with $\varphi(y) = y(1+y)^{-k}$).

$y \in I \equiv I_1 \cap I_2$ is thus necessary for a safe non-rationed equilibrium to exist, and we obtained the necessary expressions of (u, D, P). Conversely, when $y \in I$, (u, D, P) construct the unique safe non-rationed equilibrium.

Safe rationed equilibria. Compared to A.1, the only additional requirement for a token price to be compatible with a safe rationed equilibrium is that it satisfies $\lambda P\Omega \geq 1$, i.e. $P \geq \underline{P} \equiv \frac{1}{\lambda\Omega}$. The range of compatible prices is thus now $[\underline{P}, \overline{P})$, where $\overline{P}$ is given in (41). This leaves unchanged the proof that the CM auctioneer efficiently eliminates safe rationed equilibria by selecting the market-clearing price $\overline{P}$. Indeed, under Friction 2, across safe equilibria, welfare is characterized by DeFi users' utility, which increases in P across rationed equilibria as shown in A.1.

Risky non-rationed equilibria. We have $D = u\Omega$ and, from (L34), P is still given by (84). Combined with $g = \lambda P\Omega$, this implies that (L33) can be rewritten

$$(1 + uy)(1 - \alpha)\lambda \frac{(1 + y)^{-k}}{(1 - \alpha)u} = r. \tag{87}$$

Solving for u and g:

$$u = \frac{\lambda (1 + y)^{-k}}{r - \lambda y (1 + y)^{-k}}, \quad g = \frac{r - \lambda y (1 + y)^{-k}}{1 - \alpha}. \tag{88}$$

This constructs a risky non-rationed equilibrium if and only if (i) $0 \leq u \leq 1$ and (ii) $g \leq 1$.
(i) is equivalent to

$$r \geq \lambda (1 + y)^{-k+1}. \tag{89}$$

(ii) is equivalent to $r \leq 1 - \alpha + \lambda y (1 + y)^{-k}$, i.e. $y \in I_2$.

Thus, when $y \in I_2$ and (89), a unique risky non-rationed equilibrium exists. Note that in this case, a secure equilibrium also exists, because y is then also in I_1. Indeed

$$r \leq 1 - \alpha + (1 + y)^{-k+1}\lambda \frac{y}{1 + y} \leq 1 - \alpha + r\frac{y}{1 + y}.$$

The first inequality is a rewriting of $y \in I_2$, the second inequality uses (89). Reorganizing to isolate r, we obtain $r \leq (1 - \alpha)(1 + y)$, i.e. $y \in I_1$.

Risky rationed equilibria. From (L33), since $u = 1$, we obtain

$$g = \frac{r}{(1 - \alpha)(1 + y)}, \tag{90}$$

and from the specification of the security function we deduce that

$$P = \frac{1}{\lambda\Omega} \frac{r}{(1 - \alpha)(1 + y)}. \tag{91}$$

The risky rationed equilibrium exists if and only if (i) $g \leq 1$ and (ii) DeFi users do not get their ideal borrowing level: $D = \Omega < \frac{(1+y)^{-k}}{(1-\alpha)P}$. (i) is equivalent to $y \in I_1$. (ii) is equivalent to

$$r < \lambda (1 + y)^{-k+1}, \tag{92}$$

which is the complementary condition to (89). Thus, when $y \in I_1$ and (92), a unique risky rationed equilibrium exists. Note that in this case, a secure equilibrium also exists, because then $y \in I_2$. To see it, multiply both sides of (82) (which says that $y \in I_1$) by $\frac{1}{1+y}$ and both sides of (92) by $\frac{y}{1+y}$:

adding the resulting inequalities gives (85) (which means that $y \in I_2$).

We are now in a position to conclude regarding the existence and nature of equilibria. We have already seen that a risky (rationed or non-rationed) equilibrium can only exist if a safe equilibrium exists. Conversely, if $y \in I = I_1 \cap I_2$, so that a safe equilibrium exists, then one of the two complementary conditions (89) or (92) must hold, thus a risky equilibrium exists (non-rationed in the former case, rationed in the latter). This risky equilibrium is different from the safe equilibrium, except at the boundary $g = 1$, which, from the analysis above, occurs if and only if one of the two conditions $y \in I_1$ or $y \in I_2$ binds. We thus have the trichotomy:

- if $y \notin I_1$ or $y \notin I_2$: no-blockchain equilibrium;
- if $y \in I_1$ and $y \in I_2$, but not interior to both: unique blockchain equilibrium, which is secure (and non-rationed);
- if $y \in I_1$ and $y \in I_2$, interior to both: unique secure blockchain equilibrium (which is non-rationed) and unique risky blockchain equilibrium (which can be rationed or non-rationed).

In particular, when y is set at $y^{LS}_{\min} \equiv \inf I = \inf(I_1 \cap I_2)$, we are in the second case, and the equilibrium is unique (and secured, non-rationed), as announced in the second statement of the Lemma.

Our last task is to compare welfare across safe and risky equilibria when they co-exist. If the risky equilibrium is non-rationed, the utility of a given generation of DeFi users, while the blockchain operates, is the same across the two equilibria, since it is given by the same function of the same DeFi borrowing rate y. Since, obviously, the safe equilibrium has more security, it also has better welfare. Now, if the risky equilibrium is rationed, we note that the utility of DeFi users is given, as above, by (42), and to show that welfare is unambiguously improved in the safe equilibrium we need to prove that this quantity is no larger than the utility in the safe equilibrium, $\frac{1}{k-1}(1+y)^{1-k}$. This is equivalent to

$$\frac{1}{1-\frac{1}{k}}\left(\frac{1}{\lambda}\frac{r}{(1+y)}\right)^{1-\frac{1}{k}} - \frac{1}{\lambda}r \leq \frac{1}{k-1}(1+y)^{1-k} \tag{93}$$

(where we used (91)), or

$$\frac{1}{1-\frac{1}{k}}(zY^{-k})^{1-\frac{1}{k}} - zY^{1-k} \leq \frac{1}{k-1}Y^{1-k}, \tag{94}$$

where $z = \frac{1}{\lambda}r(1+y)^{k-1}$ and $Y = 1+y$. Note that $z \leq 1$ since $r < \lambda(1+y)^{-k+1}$, the condition (92) required for a risky rationed equilibrium to exist. By simplifying by $Y^{1-k} > 0$, and multiplying by $k-1$, the inequality is equivalent to

$$\begin{cases} kz^{1-\frac{1}{k}} - (k-1)z \leq 1 & \text{if } k > 1 \\ kz^{1-\frac{1}{k}} - (k-1)z \geq 1 & \text{if } k < 1. \end{cases} \tag{95}$$

This follows immediately from the analysis of the function $z \in (0;1] \mapsto kz^{1-\frac{1}{k}} - (k-1)z$.

A.9 Friction 2: Optimal DeFi rate (Corollary 1)

Using Lemma 3, for a fixed DeFi rate, welfare in the safe equilibrium is higher than in the risky equilibrium. Among safe equilibria, welfare is characterized by the utility of DeFi users, still given by (6), and is thus decreasing in y. Since $y^{LS}_{\min}$ is the smallest value such that a safe blockchain equilibrium exists, it also provides maximal DeFi users' utility, hence it maximizes welfare.

A.10 Friction 2: Proposition 4

From Carré and Gabriel (2022) (Theorem 1), we know that under direct staking, the efficient DeFi rate is

$$y^{DS}_{\min} = \frac{\rho r - 1}{u^{DS}}, \tag{96}$$

where u^{DS} is the largest solution $u \in (0;1)$ to the equation

$$\lambda = \frac{r-1}{\rho-1} u \left(1 + \frac{\rho r - 1}{u}\right)^k \tag{97}$$

if $\lambda < \frac{r-1}{\rho-1}(\rho r)^k$, and $u^{DS} = 1$ otherwise.

We first show that for sufficiently small delegation cost α, $y^{LS}_{\min} < y^{DS}_{\min}$. Given that $y^{LS}_{\min}$ is defined as the smallest rate belonging to $I_1 \cap I_2$, it is sufficient to show that $y^{DS}_{\min}$ is interior to both I_1 and I_2, i.e. that neither (82) nor (85) is binding at $y^{DS}_{\min}$ for small α. Thus, by continuity, we only need to verify that:

$$1 + y^{DS}_{\min} \quad > \quad r \tag{98}$$

$$\lambda y^{DS}_{\min}\left(1 + y^{DS}_{\min}\right)^{-k} \quad > \quad r - 1. \tag{99}$$

(98) holds because of the DS equilibrium equation $\rho r = 1 + uy$, which implies $r < \rho r = 1 + u^{DS} y^{DS}_{\min} \leq 1 + y^{DS}_{\min}$. Now, if $\lambda \geq \frac{r-1}{\rho-1}(\rho r)^k$,

$$\lambda y^{DS}_{\min}\left(1 + y^{DS}_{\min}\right)^{-k} \geq \frac{r-1}{\rho-1}(\rho r)^k (\rho r - 1)(\rho r)^{-k} > r - 1, \tag{100}$$

where we used (96) and $u^{DS} = 1$. If $\lambda < \frac{r-1}{\rho-1}(\rho r)^k$,

$$\lambda y^{DS}_{\min}\left(1 + y^{DS}_{\min}\right)^{-k} \quad = \quad \frac{r-1}{\rho-1} u^{DS}\left(1 + y^{DS}_{\min}\right)^k y^{DS}_{\min}\left(1 + y^{DS}_{\min}\right)^{-k} \tag{101}$$

$$= \quad (r-1)\frac{\rho r - 1}{\rho - 1} > r - 1, \tag{102}$$

where the first line uses (97) followed by (96) and the second line uses again $\rho r = 1 + u^{DS} y^{DS}_{min}$. Hence, (99) holds, which proves that for small α

$$y^{LS}_{\min} < y^{DS}_{\min}.$$

We now conclude the proof of the two statements of the Proposition.

(i) At $y = y^{DS}_{\min}$, under DS, the equilibrium is unique and secure. But, since $y^{LS}_{\min} < y^{DS}_{\min}$, from Lemma 3, under LS there is also a risky equilibrium (i.e. security g is strictly below 1). Thus, if the DeFi rate is fixed at $y = y^{DS}_{\min}$, liquid staking can worsen welfare, as security is 1 under DS and can be strictly below 1 in case agents coordinate on the risky LS equilibrium (this indeed leads to a strict welfare worsening, as the other welfare metrics, DeFi users' utility, is the same for LS and DS). Note that, by continuity, the same holds for DeFi rates slightly larger than $y^{DS}_{\min}$.

(ii) Adjusting optimally the DeFi rate to the chosen staking regime strictly improves DeFi users' utility, as it reduces their borrowing cost from $y_{\min}^{DS}$ to $y_{\min}^{LS}$. And it also ensures that the equilibrium is secure in both cases. Hence, liquid staking combined with an optimal adjustment of the DeFi rate strictly improves welfare. Note that no other choice of DeFi rate under LS than $y_{\min}^{LS}$ achieves this unambiguous welfare improvement.

A.11 Friction 2: Comovement of y and g across risky equilibria

Lemma 4. *Across risky non-rationed equilibria:*

— *if $k \leq 1$: security g is decreasing in the DeFi borrowing rate y;*

— *if $k > 1$: security g decreases in y as long as $y < \frac{1}{k-1}$ and increases in y thereafter.*

Across risky rationed equilibria: g decreases in y.

Intuition. k characterizes the elasticity of the DeFi users' demand: when k is small, it is inelastic, so utilization $u = \frac{D}{\Omega}$ does not react much to an increase in y. Thus, the DeFi lending rate uy increases, but it needs to be constant in risk-adjusted terms (equal to the risk-free outside option). This implies an increase in risk, i.e. a decrease in blockchain security. When k is large, DeFi users' demand is very elastic, and an increase in y thus triggers a strong decrease in u, making uy decrease and leading to the opposite conclusion.

Proof of Lemma 4. 1) Non-rationed equilibria. In the proof of Lemma 3, we have seen (eq. (88)) that

$$g = \frac{r - \lambda y (1+y)^{-k}}{1-\alpha}.$$

Hence,

$$\partial_y g = -\frac{\lambda}{1-\alpha} \left[1 + (1-k)y\right] (1+y)^{-k-1}. \tag{103}$$

There are two cases:

1. $k < 1$: then $\partial_y g$ is negative;
2. $k > 1$: then $\partial_y g$ is negative for $y < \frac{1}{k-1}$.

2) Rationed equilibria. In the proof of Lemma 3, we have seen (eq. (90)) that

$$g = \frac{r}{(1-\alpha)(1+y)}$$

so g decreases in y.

Parameters used for the Figures Figures 3, 4 and 5: $r = 1.05, \rho = 1.04, k = 2$. The first one additionally uses $\alpha = 0$ and the second one $\alpha = 3.5\%$. Figures 6, 7 and 8: $k = 1.3$, $\rho = 1.04$, $\lambda = 0.6$. The first one additionally uses $r = 1.04$ and the second one uses $\alpha = 0$.